

DOCKER

```
Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Wed Mar  5 06:31:51 2025 from 13.233.177.4
ubuntu@ip-172-31-12-45:~$ history
 1 sudo apt update
 2 sudo apt upgrade
 3 sudo apt upgrade
 4 sudo apt-get update
 5 sudo apt-get install ca-certificates curl
 6 sudo install -m 0755 -d /etc/apt/keyrings
 7 sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
 8 sudo chmod a+r /etc/apt/keyrings/docker.asc
 9 echo "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.asc] https://download.docker.com/linux/ubuntu \
10  $(. /etc/os-release && echo "${UBUNTU_CODENAME:-$VERSION_CODENAME}") stable" | sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
11 sudo apt-get update
12 sudo apt-get install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
13 sudo docker run hello-world
14 docker --version
15 docker pull python
16 sudo docker pull python
17 docker images
18 sudo su
19 history
20 docker images
21 sudo su
22 history
ubuntu@ip-172-31-12-45:~$
```

i-091f6036de0ef3285 (docker)

PublicIPs: 13.203.154.149 PrivateIPs: 172.31.12.45

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```
ubuntu@ip-172-31-12-45:~$ history
 1 sudo apt update
 2 sudo apt upgrade
 3 sudo apt upgrade
 4 sudo apt-get update
 5 sudo apt-get install ca-certificates curl
 6 sudo install -m 0755 -d /etc/apt/keyrings
 7 sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
 8 sudo chmod a+r /etc/apt/keyrings/docker.asc
 9 echo "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.asc] https://download.docker.com/linux/ubuntu \
10  $(. /etc/os-release && echo "${UBUNTU_CODENAME:-$VERSION_CODENAME}") stable" | sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
11 sudo apt-get update
12 sudo apt-get install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
13 sudo docker run hello-world
14 docker --version
15 docker pull python
16 sudo docker pull python
17 docker images
18 sudo su
19 history
20 docker images
21 sudo su
22 history
ubuntu@ip-172-31-12-45:~$ su
Password:
ubuntu@ip-172-31-12-45:~$ cd test-site
-bash: cd: test-site: No such file or directory
ubuntu@ip-172-31-12-45:~$ mkdir dev-site test-site
ubuntu@ip-172-31-12-45:~$ cd test-site
ubuntu@ip-172-31-12-45:~/test-site$ touch Dockerfile
ubuntu@ip-172-31-12-45:~/test-site$ nano Dockerfile
ubuntu@ip-172-31-12-45:~/test-site$ ls -l
total 4
-rw-rw-r-- 1 ubuntu ubuntu 15 Mar 11 07:05 Dockerfile
ubuntu@ip-172-31-12-45:~/test-site$ sudo docker build -t test-httpd .
[*] Building 6.2s (5/5) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 52B
=> [internal] load metadata for docker.io/library/httpd:2.4
docker:default
0.0s
0.0s
2.1s
```

```
ubuntu@ip-172-31-12-45:~$ test-site$ sudo docker build -t test-httpd .
[+] Building 6.2s (5/5) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile              0.0s
=> => transferring dockerfile: 52B                               0.0s
=> [internal] load metadata for docker.io/library/httpd:2.4     2.1s
=> [internal] load .dockerignore                                 0.0s
=> => transferring context: 2B                                     0.0s
=> [1/1] FROM docker.io/library/httpd:2.4@sha256:10381816bb7e60ae3a9db3784f2966a8910b6ff07c4da54bd2d62d2671c8ab6e 3.8s
=> => resolve docker.io/library/httpd:2.4@sha256:10381816bb7e60ae3a9db3784f2966a8910b6ff07c4da54bd2d62d2671c8ab6e 0.0s
=> => sha256:a0968d97f828958bcb21ffbb52261ac60b78f2275b0b10779a01f6866fee 2.10kB / 2.10kB 0.0s
=> => sha256:0de612e991352926be994158377c9f14147fe313ae92927f3b1b37dbf88ab10 7.88kB / 7.88kB 0.0s
=> => sha256:d2110b557009896789e986473b1b9100117a1301d155a969c925d79e5a8899 145B / 145B 0.5s
=> => sha256:10381816bb7e60ae3a9db3784f2966a8910b6ff07c4da54bd2d62d2671c8ab6e 10.16kB / 10.16kB 0.0s
=> => sha256:7cf63256a31a0cc4f6defe8e1af95363aee5fa75f30a248d95cae684f87c53e 28.22MB / 28.22MB 0.5s
=> => sha256:4f4fb700ef54461cfa02571ae0db9a0dc1e0cdb5577484a6d75e68dc38e8acc1 32B / 32B 0.6s
=> => sha256:38fd0d422c41e887747bb43lcc895ab880989389b3cf9212df3f0ee86dd5e640 4.20MB / 4.20MB 0.8s
=> => extracting sha256:7cf63256a31a0cc4f6defe8e1af95363aee5fa75f30a248d95cae684f87c53c 1.7s
=> => sha256:470035b3d48f5ca69f4fddcf928c47013217143ae426caa4a486e2e41ae3e21a 26.06MB / 26.06MB 1.1s
=> => sha256:fdeb6d6c6e1b28dfa478a01b758d8a3eeef5d8c457ae9c14676db5b5d607553fd 294B / 294B 0.9s
=> => extracting sha256:d2f10b557009896789e986473b1b9100117a1301d155a969c925fd79e55a8899 0.0s
=> => extracting sha256:4f4fb700ef54461cfa02571ae0db9a0dc1e0cdb5577484a6d75e68dc38e8acc1 0.0s
=> => extracting sha256:38fd0d422c41e887747bb43lcc895ab880989389b3cf9212df3f0ee86dd5e640 0.2s
=> => extracting sha256:470035b3d48f5ca69f4fddcf928c47013217143ae426caa4a486e2e41ae3e21a 0.9s
=> => extracting sha256:fdeb6d6c6e1b28dfa478a01b758d8a3eeef5d8c457ae9c14676db5b5d607553fd 0.0s
=> => exporting to image                                           0.0s
=> => exporting layers                                             0.0s
=> => writing image sha256:0de612e991352926be994158377c9f14147fe313ae92927f3b1b37dbf88ab10 0.0s
=> => naming to docker.io/library/test-httpd                      0.0s
ubuntu@ip-172-31-12-45:~$ test-site$ sudo docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
python latest 36d17f72f4f3 4 weeks ago 1.02GB
ubuntu latest a04dc4851cbc 6 weeks ago 78.1MB
test-httpd latest 0de612e99135 6 weeks ago 148MB
hello-world latest 74cc54e27dc4 6 weeks ago 10.1kB
centos latest 5d0dad3dc9764 3 years ago 231MB
ubuntu@ip-172-31-12-45:~$ test-site$ sudo docker run -dit --name test-httpd-container -p 80:80 test-httpd
35eb3caaab191cf78873ee98bb5bc5494105fb207a912220f85b03b24c749a75
ubuntu@ip-172-31-12-45:~$ test-site$ sudo docker ps -a
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
35eb3caaab19 test-httpd "httpd-foreground" 5 seconds ago Up 4 seconds 0.0.0.0:80->80/tcp, [::]:80->80/tcp test-httpd-container
centos latest 5d0dad3dc9764 3 years ago 231MB
ubuntu@ip-172-31-12-45:~$ test-site$ sudo docker run -dit --name test-httpd-container -p 80:80 test-httpd
35eb3caaab191cf78873ee98bb5bc5494105fb207a912220f85b03b24c749a75
ubuntu@ip-172-31-12-45:~$ test-site$ sudo docker ps -a
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
35eb3caaab19 test-httpd "httpd-foreground" 5 seconds ago Up 4 seconds 0.0.0.0:80->80/tcp, [::]:80->80/tcp test-httpd-container
99082f6ed976 ubuntu "/bin/bash" 6 days ago Exited (137) 6 days ago pedantic_heyrovsky
ad9acbccaf6f python "python3" 6 days ago Exited (0) 6 days ago sad_cartwright
561e1339cd21 python "python3" 6 days ago Exited (137) 6 days ago admiring_pascal
0273840f76e7 hello-world "/hello" 6 days ago Exited (0) 6 days ago focused_lovelace
ubuntu@ip-172-31-12-45:~$ test-site$ cd ~/dev-site
ubuntu@ip-172-31-12-45:~/dev-site$ mkdir public-html
cd public-html
ubuntu@ip-172-31-12-45:~/dev-site/public-html$ nano index.html
ubuntu@ip-172-31-12-45:~/dev-site/public-html$ cd ~/dev-site
ubuntu@ip-172-31-12-45:~/dev-site$ touch Dockerfile
ubuntu@ip-172-31-12-45:~/dev-site$ nano Dockerfile
ubuntu@ip-172-31-12-45:~/dev-site$ ls -l
total 8
-rw-rw-r-- 1 ubuntu ubuntu 174 Mar 11 07:07 Dockerfile
drwxrwxr-x 2 ubuntu ubuntu 4096 Mar 11 07:06 public-html
ubuntu@ip-172-31-12-45:~/dev-site$ sudo docker build -t dev-httpd .
[+] Building 1.4s (8/8) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile              0.0s
=> => transferring dockerfile: 213B                               0.0s
=> [internal] load metadata for docker.io/library/httpd:2.4     0.6s
=> [internal] load .dockerignore                                 0.0s
=> => transferring context: 2B                                     0.0s
=> [internal] load build context                                 0.0s
=> => transferring context: 185B                                    0.0s
=> CACHED [1/3] FROM docker.io/library/httpd:2.4@sha256:10381816bb7e60ae3a9db3784f2966a8910b6ff07c4da54bd2d62d2671c8ab6e 0.0s
=> [2/3] COPY ./public-html/ /usr/local/apache2/htdocs/         0.1s
=> [3/3] RUN sed -i "s/<COHORT>/408/g" /usr/local/apache2/htdocs/* 0.3s
=> => exporting to image                                           0.1s
=> => exporting layers                                             0.1s
=> => writing image sha256:e703b5c78aaa0d62c3d26ecb33783b4ddc19857e6930000c8de50610f9214dd6 0.1s
=> => naming to docker.io/library/dev-httpd                       0.0s
ubuntu@ip-172-31-12-45:~/dev-site$ sudo docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
dev-httpd latest e703b5c78aaa 3 seconds ago 148MB
crate latest ff48d7b3cb54 10 days ago 546MB
python latest 36d17f72f4f3 4 weeks ago 1.02GB
ubuntu latest a04dc4851cbc 6 weeks ago 78.1MB
test-httpd latest 0de612e99135 6 weeks ago 148MB
hello-world latest 74cc54e27dc4 6 weeks ago 10.1kB
centos latest 5d0dad3dc9764 3 years ago 231MB
ubuntu@ip-172-31-12-45:~/dev-site$ sudo docker run -dit --name dev-httpd-container -p 8080:80 dev-httpd
2a8f397730013276715741d548b09f124e28f2040d7dc3c3d3834929c0e865cc
ubuntu@ip-172-31-12-45:~/dev-site$ sudo docker ps -a
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
2a8f39773001 dev-httpd "httpd-foreground" 4 seconds ago Up 3 seconds 0.0.0.0:8080->80/tcp, [::]:8080->80/tcp dev-httpd-container
35eb3caaab19 test-httpd "httpd-foreground" 2 minutes ago Up 2 minutes 0.0.0.0:80->80/tcp, [::]:80->80/tcp test-httpd-container
99082f6ed976 ubuntu "/bin/bash" 6 days ago Exited (137) 6 days ago pedantic_heyrovsky
ad9acbccaf6f python "python3" 6 days ago Exited (0) 6 days ago sad_cartwright
561e1339cd21 python "python3" 6 days ago Exited (137) 6 days ago admiring_pascal
0273840f76e7 hello-world "/hello" 6 days ago Exited (0) 6 days ago focused_lovelace
ubuntu@ip-172-31-12-45:~/dev-site$
```

JENKINS

Dashboard > My-First-Job > Configuration

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps**
- Post-build Actions

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute shell ?

Command

See the list of available environment variables

```
echo "🚀 Hello, Jenkins!"
echo "Today's Date: ${date}"
echo "This is my first Jenkins Job!"
```

Advanced ▾

Add build step ▾

Post-build Actions

Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.

Add post-build action ▾

Save Apply

- Status
- Changes
- Console Output**
- Edit Build Information
- Delete build '#2'
- Timings
- Previous Build

Console Output

Download Copy View as plain text

```
Started by user Bhanu Prakash
Running as SYSTEM
Building in workspace /Users/bhanu.y/.jenkins/workspace/My-First-Job
[My-First-Job] $ /bin/sh -xe /var/folders/g/_/j8_jqnds6fs5lgtgsz30lt3h0000gn/T/jenkins14742361135601817198.sh
+ echo '🚀 Hello, Jenkins!'
🚀 Hello, Jenkins!
++ date
+ echo 'Today'\''s Date: Tue Mar  4 10:00:21 IST 2025'
Today's Date: Tue Mar  4 10:00:21 IST 2025
+ echo 'This is my first Jenkins Job!'
This is my first Jenkins Job!
Finished: SUCCESS
```

Dashboard > hw > #1

Status

Changes

Console Output

Edit Build Information

Delete build '#1'

Timings

Pipeline Overview

Pipeline Console

Replay

Pipeline Steps

Workspaces

Next Build

✓ #1 (3 Mar 2025, 10:39:31)

Add description

Keep this build forever

Started by user [Bhanu Prakash](#)

Started 7 days 20 hr ago
Took [4.5 sec](#)

This run spent:

- 49 ms waiting;
- 4.5 sec build duration;
- 4.5 sec total from scheduled to completion.

</>

No changes.

REST APIJenkins 2.492.1

C408 Foundation of SRE: Kul x devops - Google Docs x Instances | EC2 | ap-south-1 x hw #1 Console [Jenkins] x EC2 Instance Connect | ap-s x +

← → ↻ 15.207.111.124:8080/job/hw/1/console

Dashboard > hw > #1

Timings

Pipeline Overview

Pipeline Console

Restart from Stage

Replay

Pipeline Steps

Workspaces

Next Build

```
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Birth)
[Pipeline] echo
Welcome to the world!
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Education)
[Pipeline] echo
Learn, grow, and develop new skills.
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Career)
[Pipeline] echo
Find your passion and pursue a fulfilling career.
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Retirement)
[Pipeline] echo
Relax, enjoy the fruits of your labor, and reflect on a life well-lived.
[Pipeline] }
[Pipeline] // stage
[Pipeline] End of Pipeline
Finished: SUCCESS
```

REST APIJenkins 2.492.1

Configure

- General
- Triggers
- Pipeline
- Advanced

Definition

Pipeline script

Script ?

```
1- pipeline {
2-   agent none
3-   stages {
4-     stage('Hello') {
5-       steps {
6-         echo 'Hello World'
7-       }
8-     }
9-     stage('Birth') {
10-      steps {
11-        echo 'Welcome to the world!'
12-      }
13-    }
14-    stage('Education') {
15-      steps {
16-        echo 'Learn, grow, and develop new skills.'
17-      }
18-    }
19-    stage('Career') {
20-      steps {
21-        echo 'Find your passion and pursue a fulfilling career.'
22-      }
23-    }
24-    stage('Retirement') {
25-      steps {
26-        echo 'Relax, enjoy the fruits of your labor, and reflect on a life well-lived.'
27-      }
28-    }
29-  }
}
```

☒ Use Groovy Sandbox ?

[Pipeline Syntax](#)

Save

Apply

Browser tabs: C408 Foundation of SRE: Ku..., devops - Google Docs, Instances | EC2 | ap-south-1, hw #1 Console [Jenkins], EC2 Instance Connect | ap-s...

Address bar: Not Secure 15.207.111.124:8080/job/hw/1/console

Jenkins logo and navigation icons

Dashboard > hw > #1

Status

Changes

Console Output

Edit Build Information

Delete build '#1'

Timings

Pipeline Overview

Pipeline Console

Restart from Stage

Replay

Pipeline Steps

Workspaces

Next Build

Console Output

Download

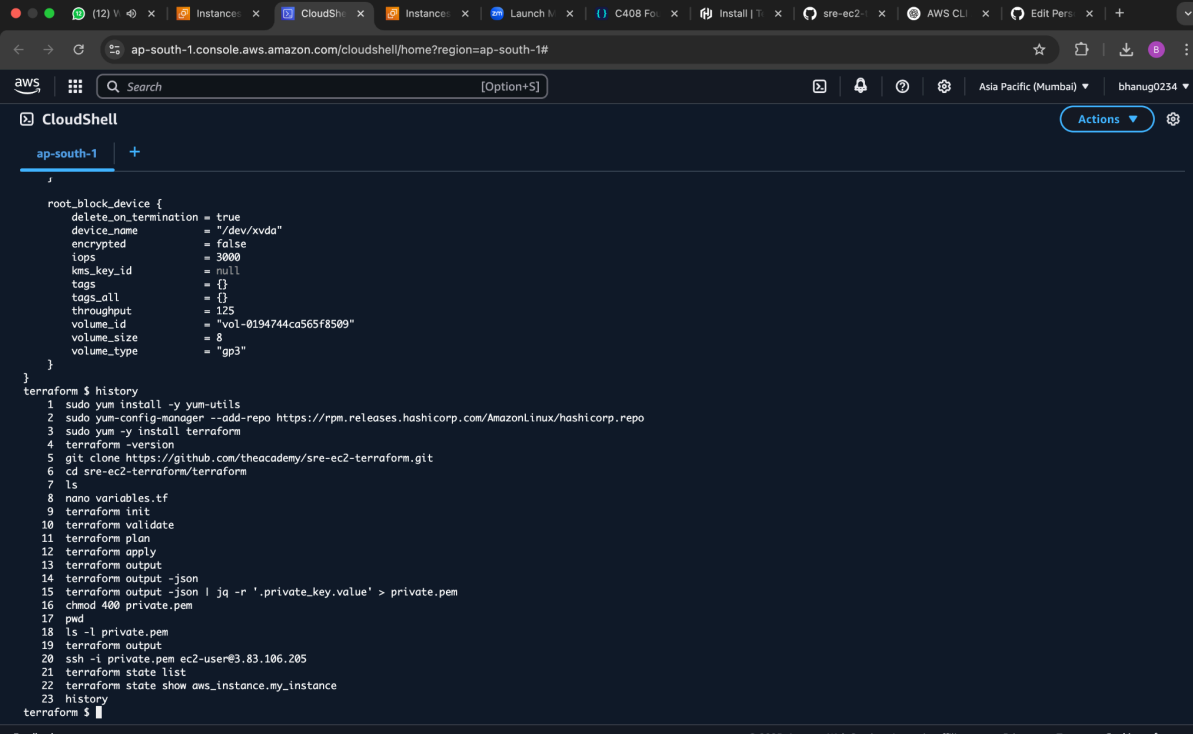
Copy

View as plain text

Started by user [Bhanu Prakash](#)

```
[Pipeline] Start of Pipeline
[Pipeline] stage
[Pipeline] { (Hello)
[Pipeline] echo
Hello World
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Birth)
[Pipeline] echo
Welcome to the world!
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Education)
[Pipeline] echo
Learn, grow, and develop new skills.
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Career)
[Pipeline] echo
Find your passion and pursue a fulfilling career.
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
```

TERRAFORM

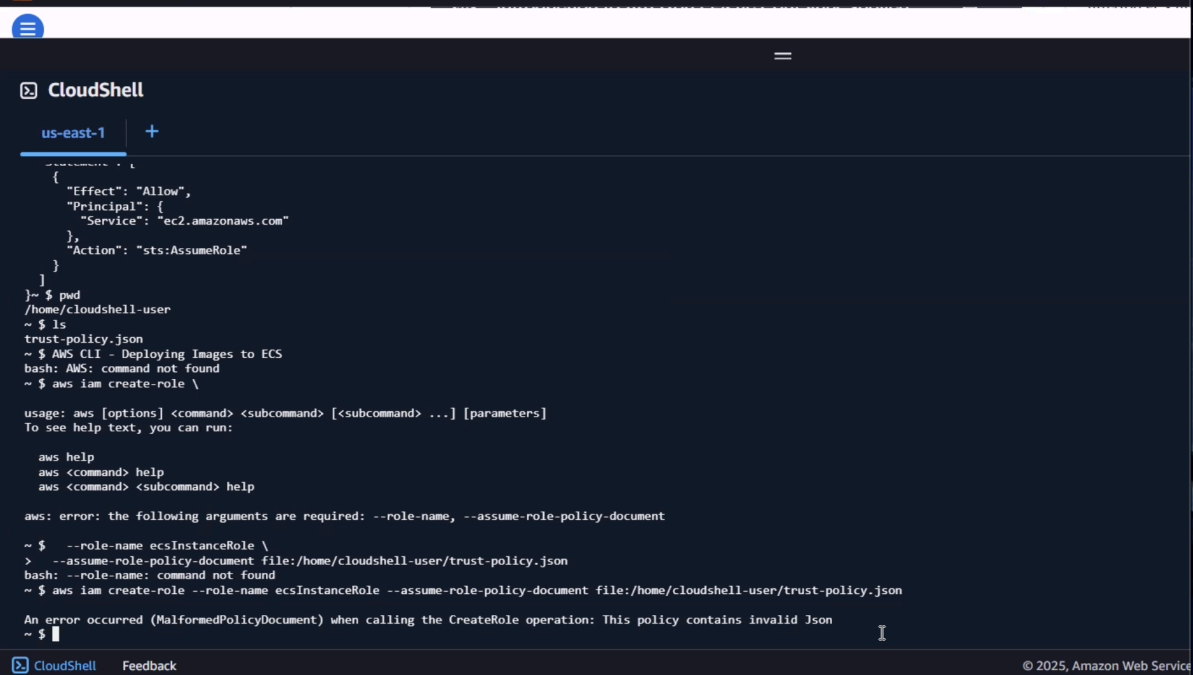


The screenshot shows the AWS CloudShell interface for the 'ap-south-1' region. The terminal displays the following commands and output:

```
root_block_device {
  delete_on_termination = true
  device_name           = "/dev/xvda"
  encrypted              = false
  iops                   = 3000
  kms_key_id             = null
  tags                  = {}
  throughput             = 125
  volume_id              = "vol-0194744ca565f8509"
  volume_size            = 8
  volume_type            = "gp3"
}
}
terraform $ history
1 sudo yum install -y yum-utils
2 sudo yum-config-manager --add-repo https://rpm.releases.hashiCorp.com/AmazonLinux/hashicorp.repo
3 sudo yum -y install terraform
4 terraform -version
5 git clone https://github.com/theacademy/sre-ec2-terraform.git
6 cd sre-ec2-terraform/terraform
7 ls
8 nano variables.tf
9 terraform init
10 terraform validate
11 terraform plan
12 terraform apply
13 terraform output
14 terraform output -json
15 terraform output -json | jq -r '.private_key.value' > private.pem
16 chmod 400 private.pem
17 pwd
18 ls -l private.pem
19 terraform output
20 ssh -i private.pem ec2-user@83.83.106.205
21 terraform state list
22 terraform state show aws_instance.my_instance
23 history
terraform $
```

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Aws cli



The screenshot shows the AWS CloudShell interface for the 'us-east-1' region. The terminal displays the following commands and output:

```
{
  "Effect": "Allow",
  "Principal": {
    "Service": "ec2.amazonaws.com"
  },
  "Action": "sts:AssumeRole"
}
}
~$ pwd
/home/cloudshell-user
~$ ls
trust-policy.json
~$ AWS CLI - Deploying Images to ECS
bash: AWS: command not found
~$ aws iam create-role \

usage: aws [options] <command> [<subcommand> ...] [parameters]
To see help text, you can run:

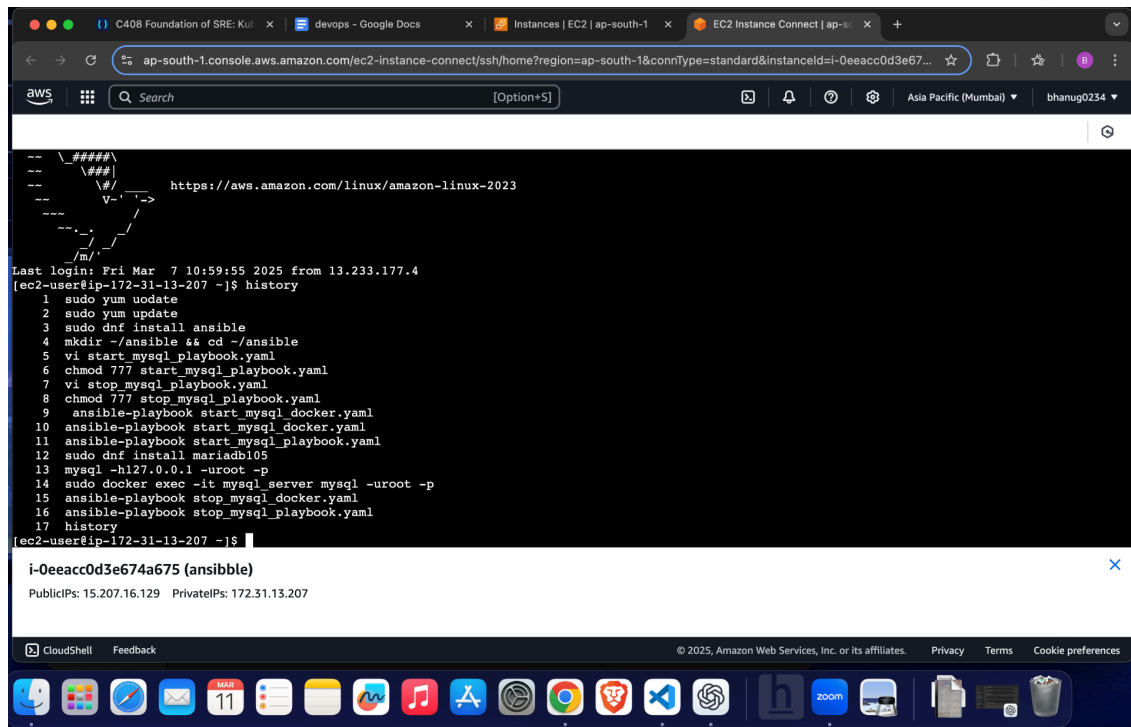
    aws help
    aws <command> help
    aws <command> <subcommand> help

aws: error: the following arguments are required: --role-name, --assume-role-policy-document
~$ --role-name ecsInstanceRole \
> --assume-role-policy-document file:/home/cloudshell-user/trust-policy.json
bash: --role-name: command not found
~$ aws iam create-role --role-name ecsInstanceRole --assume-role-policy-document file:/home/cloudshell-user/trust-policy.json

An error occurred (MalformedPolicyDocument) when calling the CreateRole operation: This policy contains invalid json
~$
```

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Ansible



K8s

[illegible]

